EXPERIMENT 10

Performance Optimization in Zoho Creator

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Subject: Low Code Application Development using Zoho Creator

AIM

To demonstrate performance optimization techniques in the **Student Admission** application in Zoho Creator, particularly efficient record retrieval and workflow processing.

INTRODUCTION

Performance optimization in low-code applications means designing forms, workflows, and scripts so that they use data and processing only as far as needed. Although a low-code platform handles much of the underlying work, the way in which a workflow is written still influences how efficiently it runs.

Retrieving more records than required, or repeatedly processing records that are not relevant to the task, causes unnecessary data retrieval and processing. In a small application this may not be noticeable, but as the number of records and users grows, it can reduce application efficiency. Efficient workflows in Zoho Creator therefore retrieve only the data they need and avoid unnecessary processing, which supports scalable and maintainable applications.

APPLICATION USED

The practical was performed in Zoho Creator using the **Student Admission** application. Deluge scripting was used to write the logic of a form workflow that sends an email notification when an admission registration is submitted. No upgrade-only feature is claimed as enabled or configured in this record.

PERFORMANCE OPTIMIZATION CONCEPT

The key optimization demonstrated is criteria-based record fetching. Instead of retrieving every record of a form, a criteria is given so that only the required record is returned. In the Student Admission workflow, this is written as:

```
studentAdmission = Student_Admission [ID == input.ID];
```

- The workflow identifies the current record using `input.ID`, which holds the ID of the record being processed.
- Only the relevant Student Admission record, the one whose ID matches `input.ID`, is retrieved.
- This avoids unnecessarily fetching and processing unrelated records.
- This improves workflow efficiency and reduces unnecessary data processing.

PROCEDURE

1. Open the **Student Admission** application in Zoho Creator in Edit mode.
2. Navigate to the Workflow section and open Form workflows. The form **Student Admission** is listed under Forms with workflows.
3. Observe the form workflow "Send admission confirmation email". It is triggered on Created > Successful form submission, has 1 action, and has the status Enabled (Figure 1).

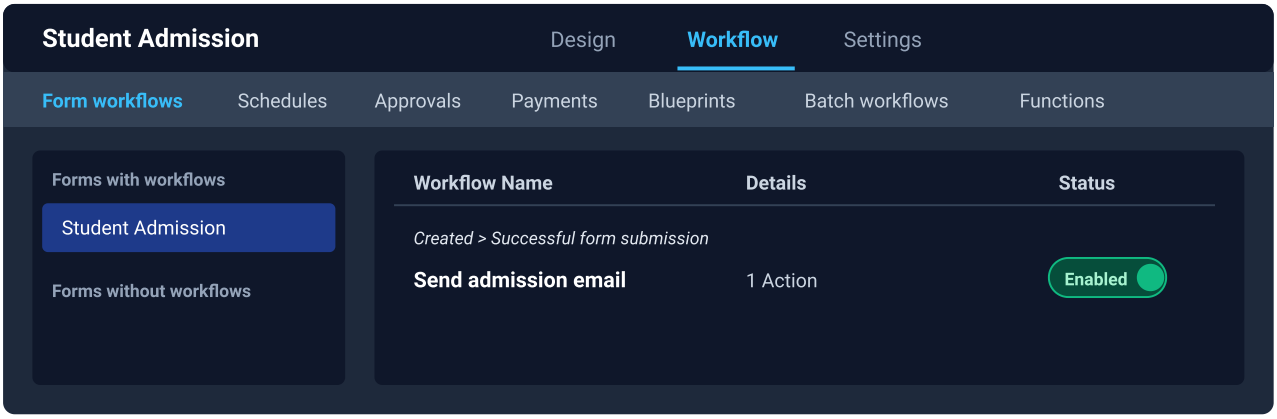


Figure 1: Form Workflow of the Student Admission Application (Illustrative Representation)

4. When a Student Admission form is submitted, the workflow identifies the current record using `input.ID`.
5. The required record is fetched using the criteria `ID == input.ID`, so that only this record is retrieved.
6. The workflow checks whether the email field of the record is blank.
7. If the email is available, the notification email is sent to the applicant. If the email is missing, an informational message is displayed.
8. The Deluge script used in the workflow is given in the next section.

DELUGE CODE

The Deluge code of the workflow is as follows:

```
studentAdmission = Student_Admission[ID == input.ID]; if(studentAdmission.email.isBlank()) { info "Email address is missing for this registration"; } else { sendmail [ from : zoho.adminuserid to : studentAdmission.email subject : "Your Student Admission is Confirmed" message : "<p>Dear Applicant,</p><p>Your admission application for the academic year has been successfully received.</p><p>Thank you for applying.</p>" ] }
```

The first statement fetches the record by its ID. If the email field of that record is blank, an informational message is displayed. Otherwise, an email with the subject *"Your Student Admission is Confirmed"* is sent to the applicant's email address.

PERFORMANCE ANALYSIS

The optimized approach is compared with a general inefficient approach below.

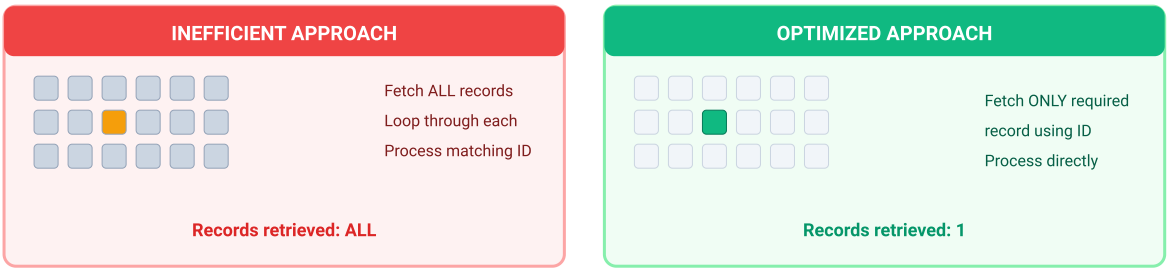


Figure 2: Records Retrieved by Inefficient vs Optimized Approaches (Conceptual Diagram)

Illustrative comparison only (inefficient code not used in the application):

```
studentAdmissions = Student_Admission[]; for each rec in studentAdmissions { if (rec.ID == input.ID) { // Process record... } }
```

No benchmark was performed in this practical, so no measured improvement in execution time is claimed. The comparison is based on the amount of data retrieved and processed.

OPTIMIZATION TECHNIQUES

General practices that contribute to performance in Zoho Creator applications are given below and shown in Figure 3, a conceptual diagram.

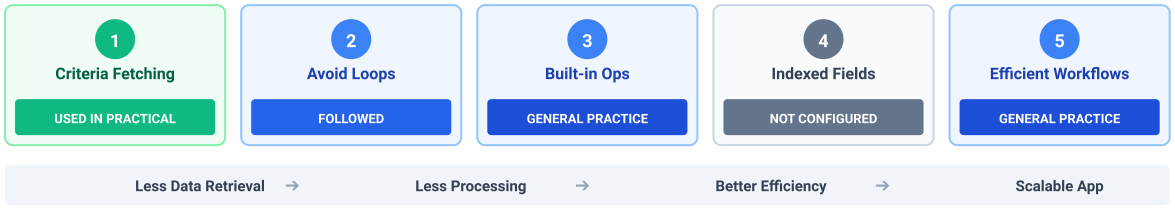


Figure 3: Performance Optimization Techniques (Conceptual Diagram)

Technique	Explanation
Criteria-based record fetching	Retrieve only the records required for the current operation. This technique is used in the practical.
Avoiding unnecessary record iteration	Do not fetch all records and loop through them when a specific criterion can be used.
Efficient built-in functions	Use built-in functions and aggregate operations instead of unnecessary repeated processing.
Indexed fields	Indexed fields can improve retrieval performance for frequently queried data. Indexing was not enabled or configured in this application.
Efficient workflows	Keep workflows simple and avoid unnecessary nested or repeated operations.

OPTIMIZED WORKFLOW FLOWCHART

Figure 4 is a conceptual flowchart of the optimized Student Admission workflow:

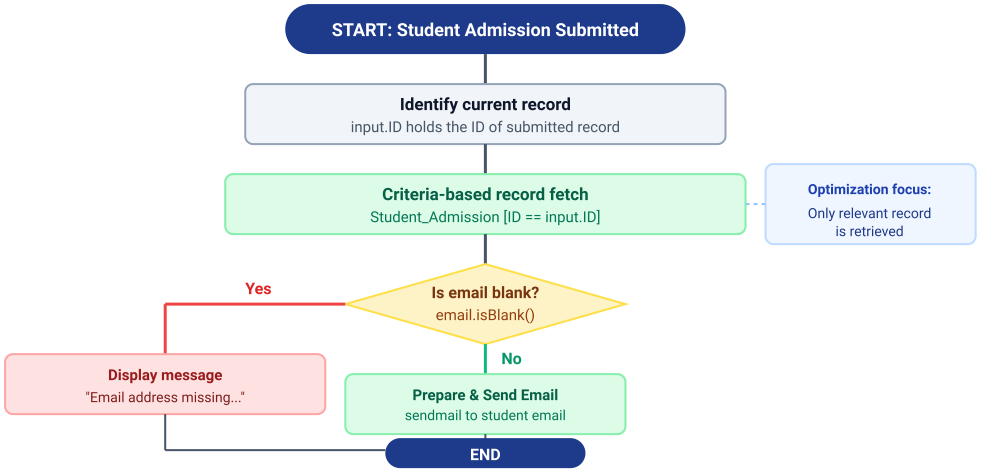


Figure 4: Optimized Student Admission Workflow (Conceptual Flowchart)

1. **Identify current record:** `input.ID` gives the ID of the submitted record.
2. **Criteria-based record fetch:** `Student_Admission [ID == input.ID]` retrieves only the relevant record.
3. **Validate email:** `email.isBlank()` checks whether the email field is empty.
4. **Prepare notification and send email:** If the email is available, the notification message is prepared and sent using `sendmail`; otherwise an informational message is displayed.

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CONCLUSION

Performance optimization is important in low-code applications because efficiency depends not only on the platform but also on how workflows and scripts are written. In this practical, the **Student Admission** workflow used criteria-based record retrieval with the expression `Student_Admission [ID == input.ID]`, so that only the required record was retrieved and processed.

Targeted record retrieval improves efficiency by avoiding the retrieval of unrelated records and the iteration through them, and efficient workflow design reduces unnecessary processing. Practices such as criteria-based fetching, avoiding unnecessary loops, using built-in operations, and keeping workflows simple help applications to handle growing amounts of data. These practices contribute to scalable and maintainable Zoho Creator applications. No benchmark was performed, so the benefits are described in terms of reduced data retrieval and processing and not as measured execution times.

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